

Microcontroller Solutions

Brian McAllister
Director, Marketing
Development Systems Division

1906

- World's best selling toy



Original STEIFF Bear
circa 1903

2006

- Toy manufacturers recognise the amazing versatility of 32-bit processors and implement them in all sorts of wonderful designs



LEGO Mindstorm NXT
ARM7™ family-based MCU

Welcome to
The World's Biggest LEGO® Shop!

▶ 4894
**Mythical
Creatures**

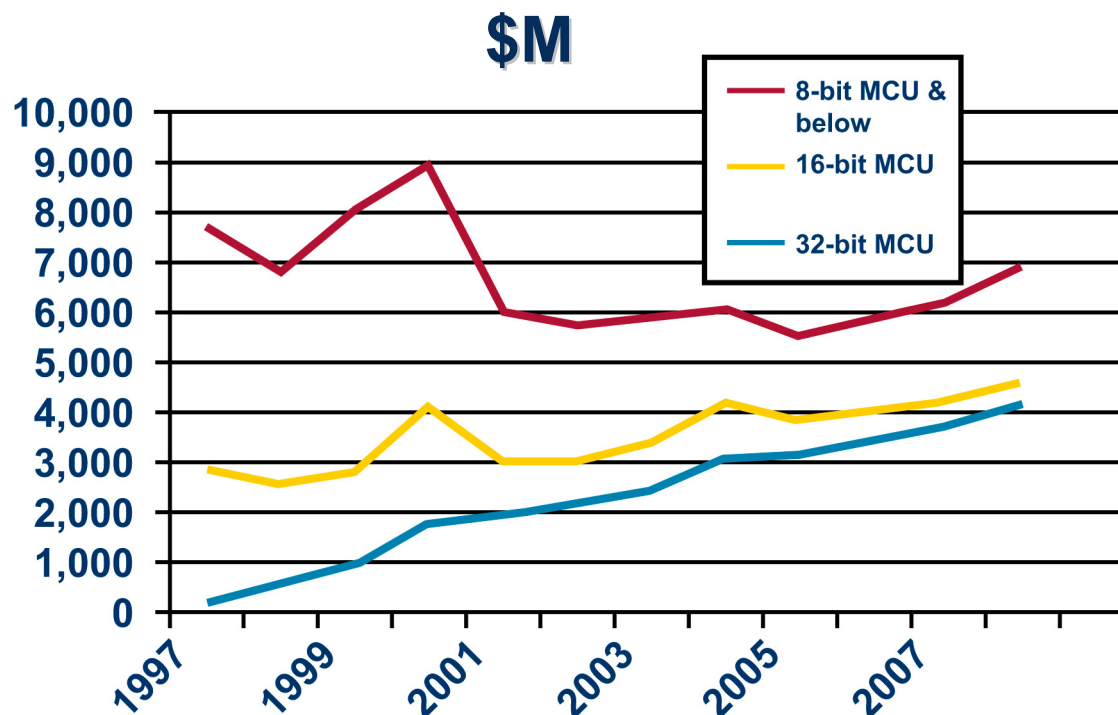
NEW!

**What's Hot Now
Top 10 Sellers:**

- 1 [LEGO® MINDSTORMS® NXT](#)
- 2 [Jabba's Sail Barge™](#)
- 3 [My Custom Design](#)

Next-Generation Microcontrollers

- Base MCU platform becoming a commodity
- Differentiation through integration
- Speed-to-market is the key to obtaining a competitive edge
 - Example: Metro™ enabling ultra low power UWB microcontrollers



ARM in MCUs Today

- Significant success to date
 - ~70M units shipped in 2005
 - More than 100 ARM technology-based MCUs in market today
 - ARM7 family-based MCUs are selling for below \$2
 - More than 30,000 members of ARM processor-based MCU user groups
- ARM7 family forms the basis of a great microcontroller
- ARM9™ family used for performance or Embedded Linux designs

"In North America, we have seen a **30% increase in ARM based design** activity over the last 18 months" - David Doherty, Vice President - Arrow North American Components

MCU Families

	ARM7TDMI	ARM9
 ANALOG DEVICES	ADuC7000	
 ATMEL	AT91SAM7x	AT91SAM9x
 Launched by Motorola freescalse semiconductor	MAC7xxx	
 OKI	MLQ2xx	ML69xx
 PHILIPS	LPC2xx	LPC3xx
 SHARP	LH7A404	
 ST	STR7	STR9
 TEXAS INSTRUMENTS	TMS470	
 Actel®	MP7 (FPGA)	

32-bit MCU Cheaper Than McDonald's



M I C R O P R O C E S S O R

www.MPRonline.com

THE INSIDER'S GUIDE TO MICROPROCESSOR HARDWARE

PHILIPS CHALLENGES 8-BIT MCUs

New 32-Bit ARM7 Microcontrollers With Flash Memory Start at \$1.47

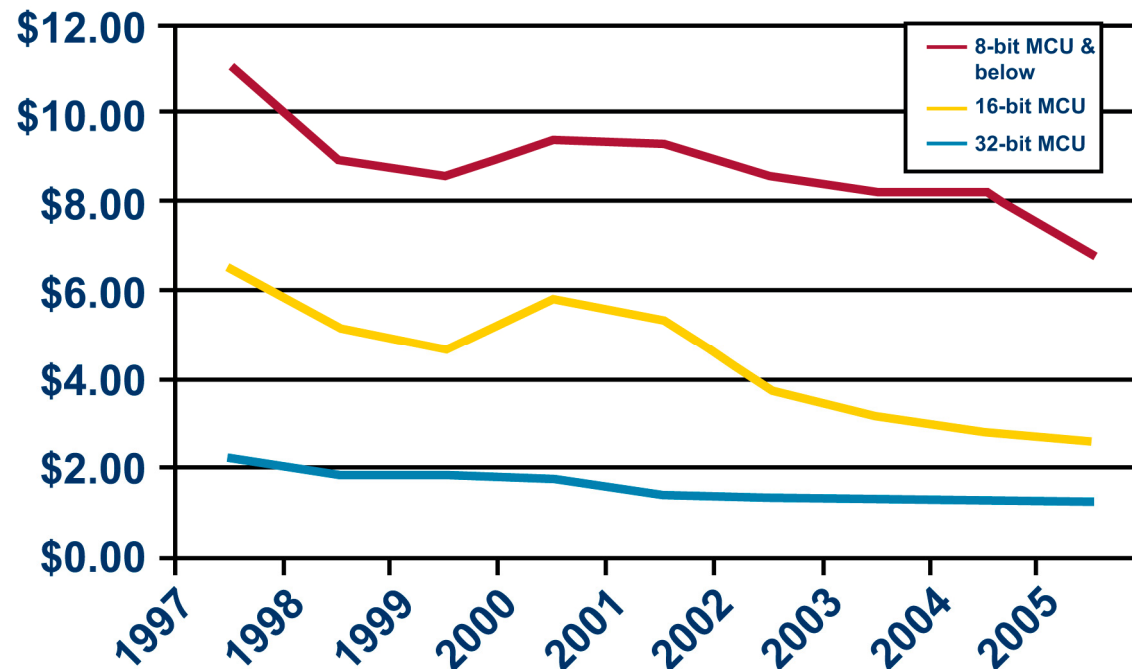
By Tom R. Halfhill {10/10/05-01}

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If you have \$1.50 in pocket change, you can buy two quarts of gasoline, a Big Mac with cheese at McDonald's, or a 32-bit microcontroller with a 70MHz ARM7 processor and 8KB of internal flash memory. The choice is clear.

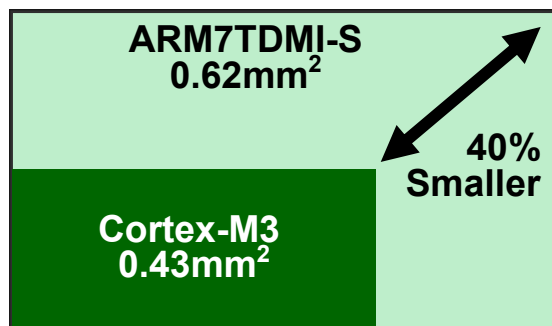
ARM Cortex™-M3 is Now Shipping

- ARM processors now range from \$1 to 1 GHz
- ARM Cortex-M3 production release available December 2005
- First MCU's released on March 27th 2006
- All supported by a single compiler suite



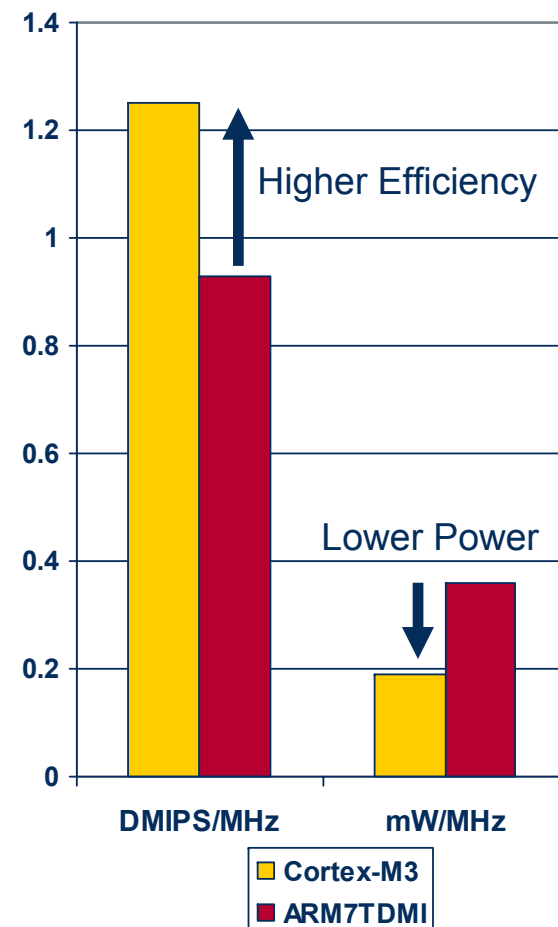
Evolution for Success

- The Cortex-M3 processor delivers many advantages for future applications:-
 - Silicon costs reduced by in excess 27%
 - Processor area reduced by up to 40%
 - Memory requirements reduced by up to 30%
 - Reduced frequency requirement of up to 40%
 - Power consumption reduced by up to 65%
 - Performance increase of up to 60%



0.18u TSMC Generic

Note: Advantages are based on user choices and may not all be achievable within the same application



How Does 32-bit Cost Compare ?

■ Costing Example

- Performance Requirement: **5 DMIPS**
- Code Size: **256KB equivalent code**
- Process: TSMC 0.18 μ Generic



	Cortex-M3	8051
Frequency Requirement	4MHz (5DMIPS / 1.25DMIPS/MHz)	63MHz (5DMIPS / 0.08DMIPS/MHz)
Power Consumption	0.8mW (0.19mW/MHz x 4MHz)	32mW (0.5mW/MHz x 63MHz)
Core Silicon Area	0.36mm²	0.10mm² (approx.)
Flash Memory Area	0.13mm² (4x Code Density = 64KB)	0.52mm² (256KB)
Total Area (Core+Flash)	0.49mm²	0.62mm²
Basic Cost of Silicon*	19c per die	20c per die

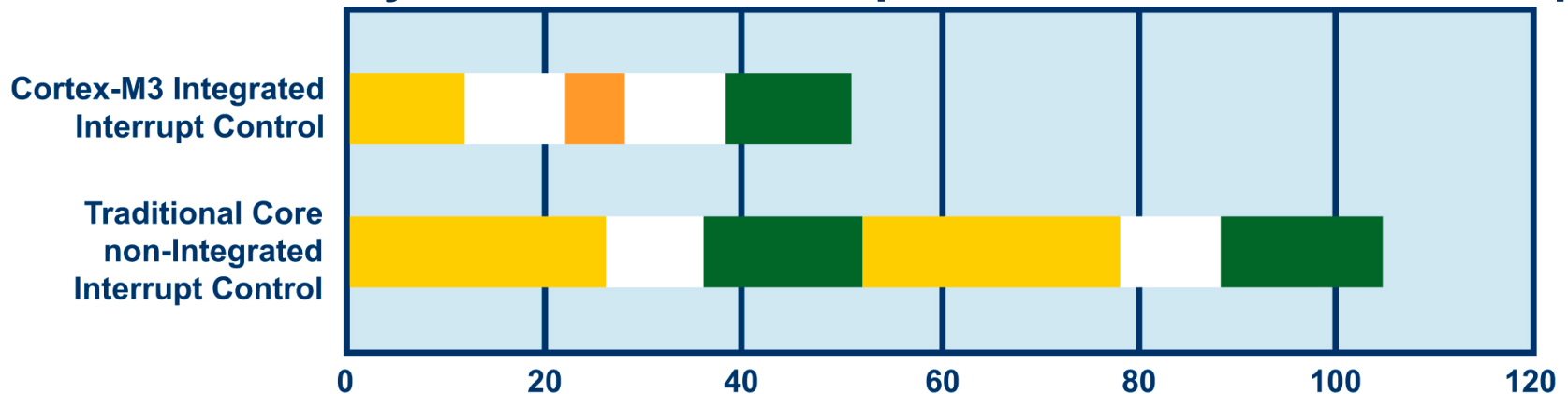
ARM Cortex-M3 processor delivers more than 15x performance and 95% better power efficiency for the same silicon cost

*Costs calculated by IC Knowledge 2005

ARM Cortex-M3 Processor Lowers Cost

- **Clock Frequency:** Cycles to complete critical interrupts dictates MHz. Cortex-M3 processor enters and handles interrupts in fewer cycles.

Cycles needed to complete 2 'back-to-back' interrupts



- **RAM optimization** - unaligned memory access allowing 8- /16- /32-bit values at any location
- **Low cost debug tools solution** enabled by sophisticated on-chip logic for Breakpoint, Data Watch, Event Trigger and Profiling

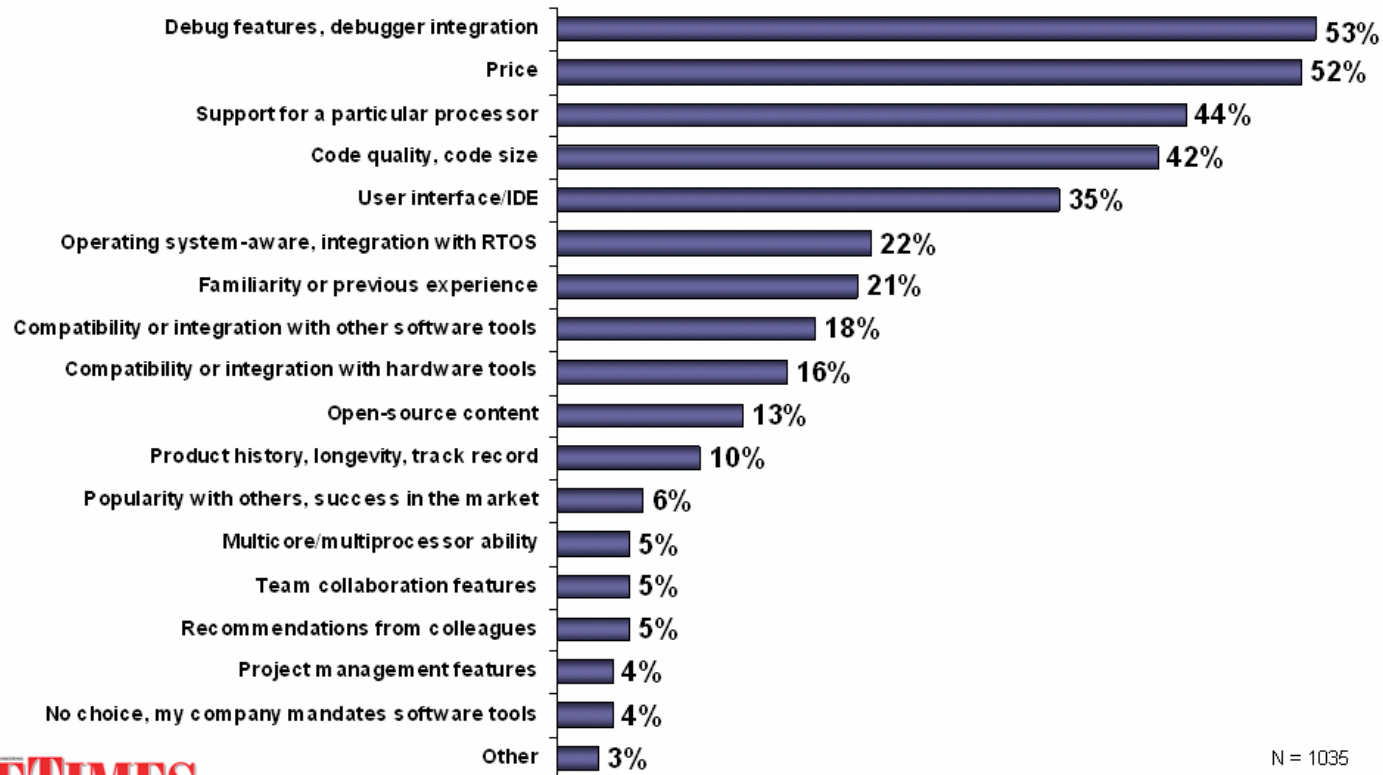
“Silicon Without Software is Just Sand”



27

2006 State of Embedded Market Survey

Software tool selection factors



N = 1035

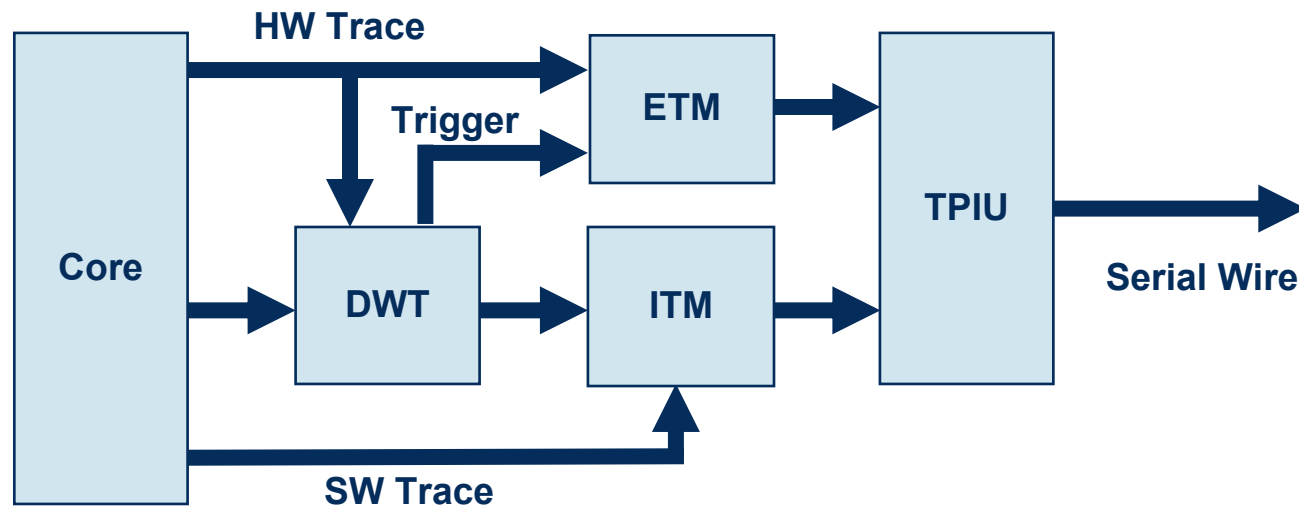


22. What are the most important factors in choosing software tools?

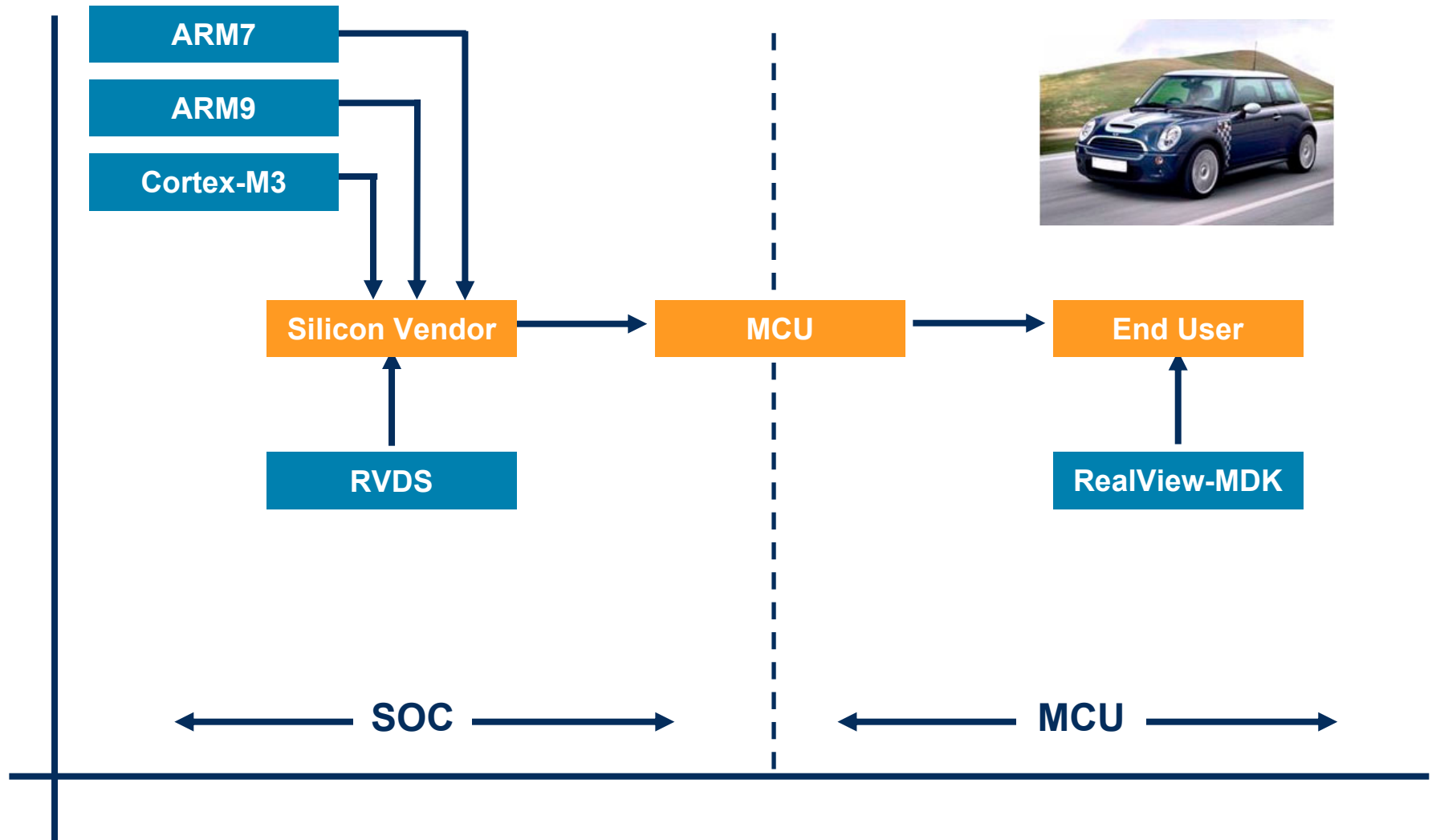


Cortex-M3 Processor - Debug Friendly

- Real-time trace using the Embedded Trace Macrocell™ (ETM)
- ETM provides real-time trace of instruction execution and data transfers
- ETM triggers on specific transfers and instruction and data watch points
- Serial Wire debug enables low pin count



MCU Development Process



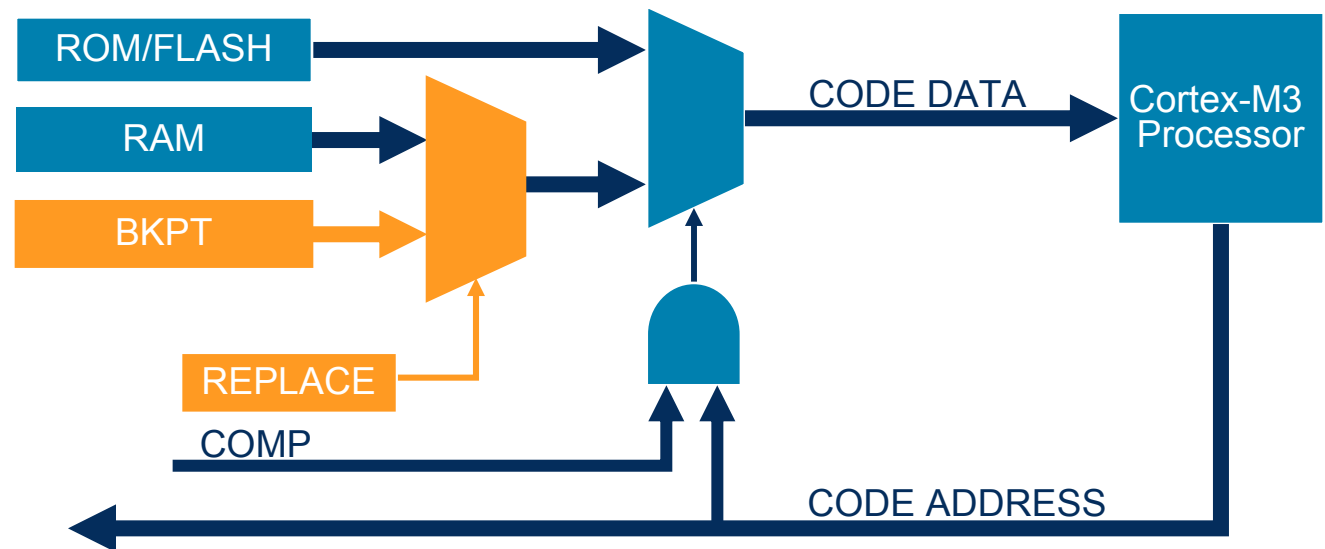
RealView® Tools for Device Creation

- RealView Development Suite 3.0 offers full support for the Cortex-M3 processor
 - Includes compiler, debugger, Eclipse IDE
 - Support for Thumb®-2
 - Also includes high-performance simulator
- Full on chip debug support available with RealView ICE
- Emulator baseboard with FPGA images for fast hardware system prototyping now available



Flash Patch and Breakpoint Unit

- Allows debug actions to be performed on read-only code
 - Independent remapping of ROM code and data to RAM region
 - Configurable insertion of software breakpoints on address match
 - Writes are never patched
 - Low cost, non-intrusive, code profiling solution
 - Equivalent to having eight hardware breakpoints



Keil™ an ARM Company

- Leading supplier of MCU development tools

- ANSI C/C++ compilers, debuggers.
- Device simulators
- Real-time kernels, real-time libraries.

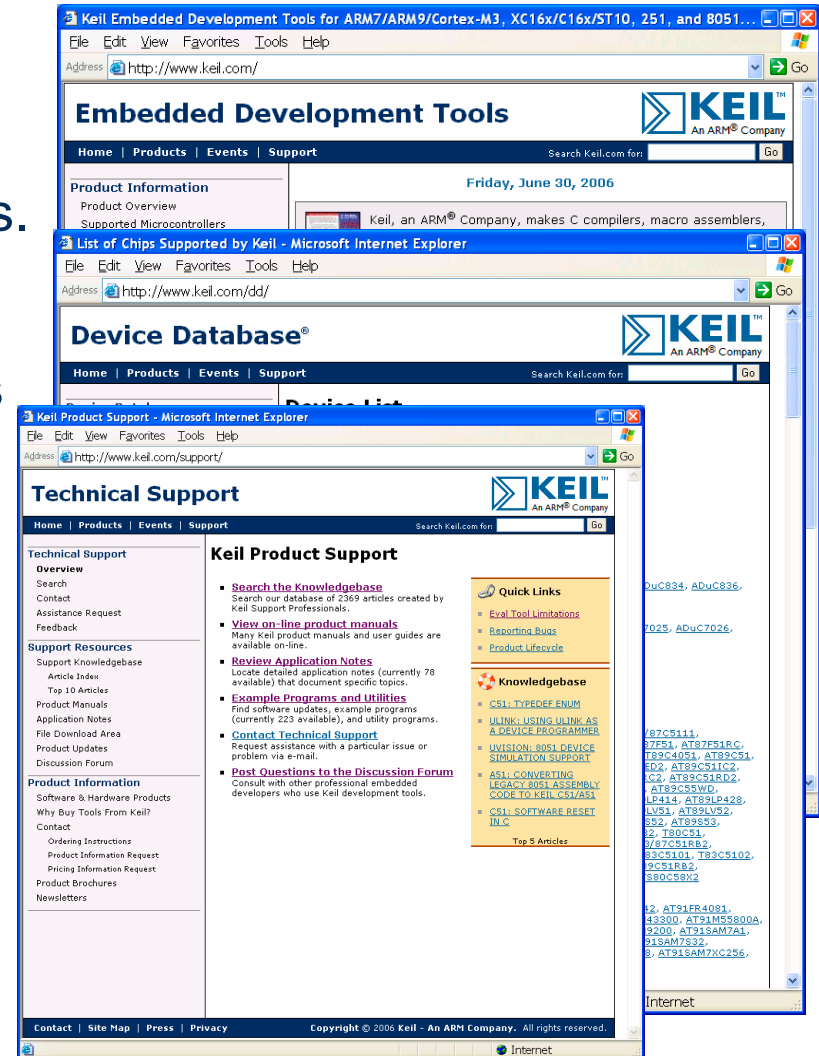
- Extensive device database

- Including >1000 8-/16-/32-bit MCUs

- Excellent, established support

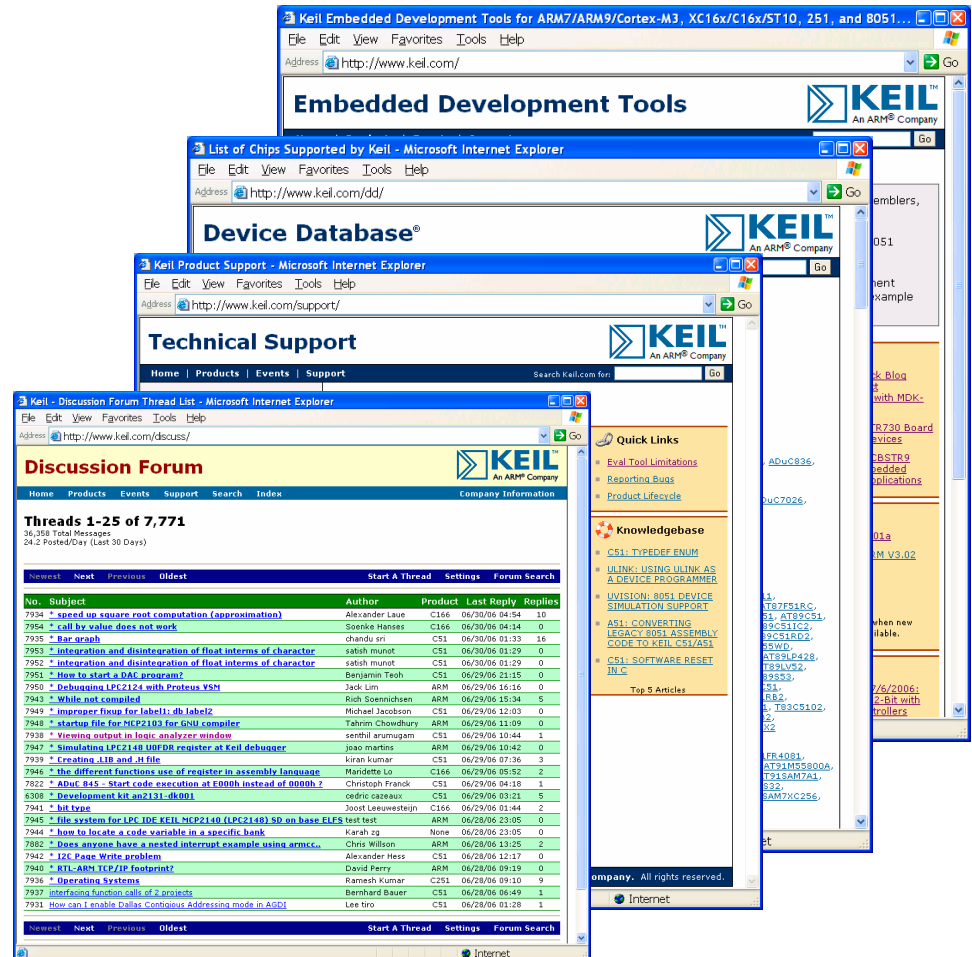
- Web support portal
- User group structure
- Global distribution network

- 100k+ users



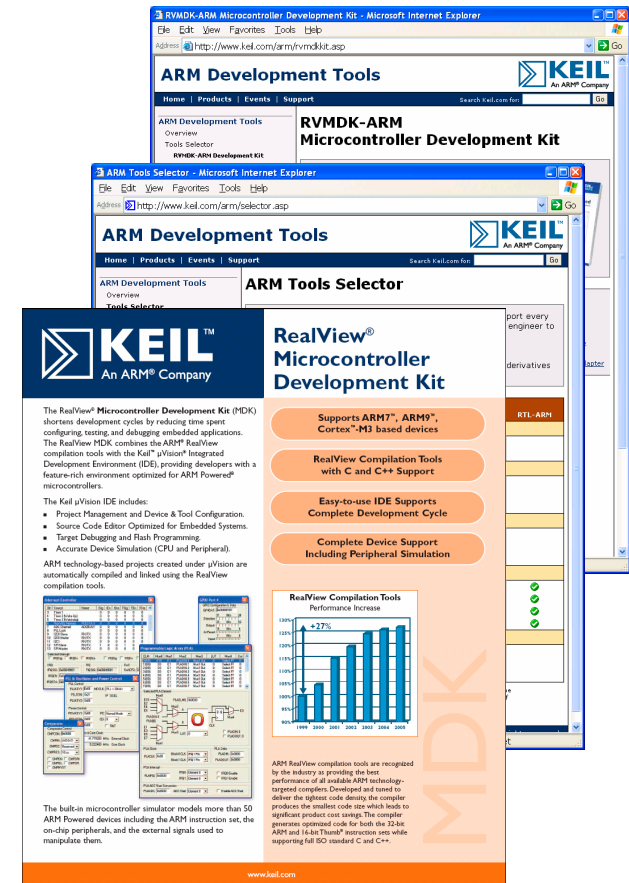
Mass Market Support

- Application notes
- Program examples
- Device database
- Support knowledge
- Discussion forum



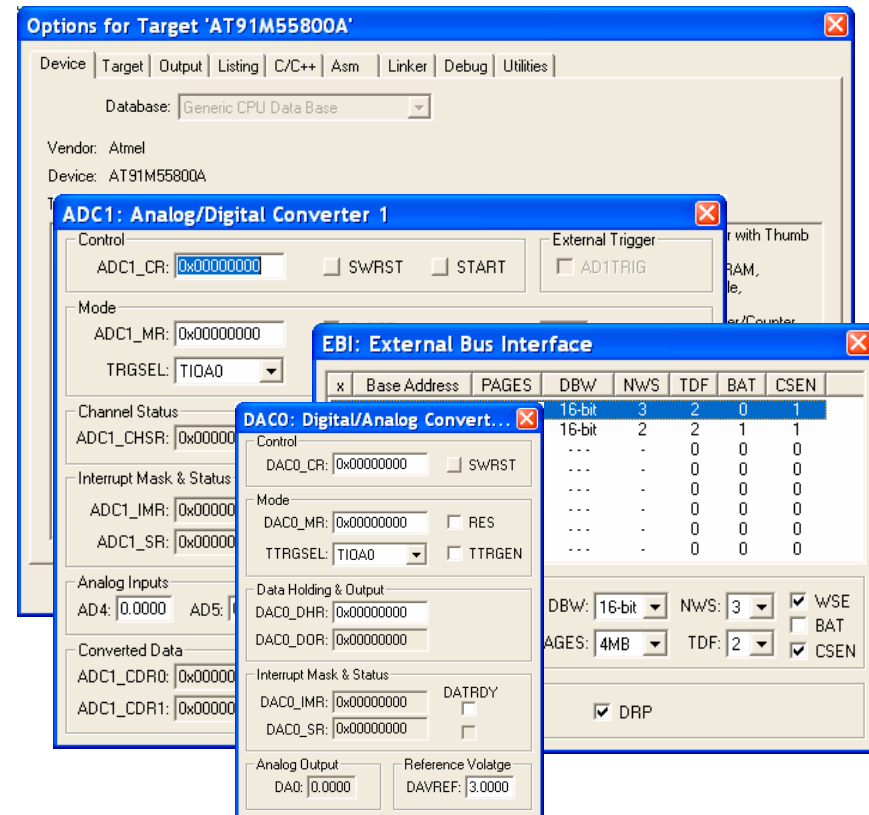
RealView Microcontroller Development Kit

- Industry-leading technology
 - RealView compilation tools
 - Keil μ Vision Integrated Development Environment
- Supports ARM7, ARM9 and Cortex-M3 processor-based MCUs
- Optional Hardware Products
 - ULINK USB-JTAG adapter
 - Evaluation boards
- RTX Real Time Kernel
 - Efficient RTOS Kernel for small systems



Microcontroller Support

- Out of the box support
 - >150 ARM MCU's.
 - See www.keil.com/dd
- Complete simulation of target
 - Simulates entire target hardware
 - Including external signals and I/O
 - Fast Instruction Set Simulation
 - Interrupt simulation
 - On-chip peripheral's
 - ADC, DAC, EBI, Timers
 - UART, CAN, I2C



End Users Can Create Program

Step 1: Select Device & Specify Target Hardware

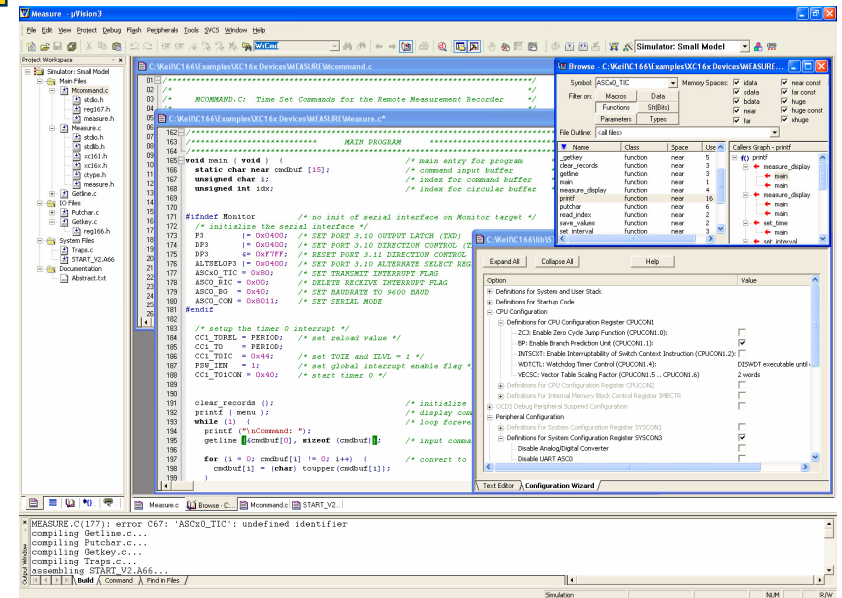
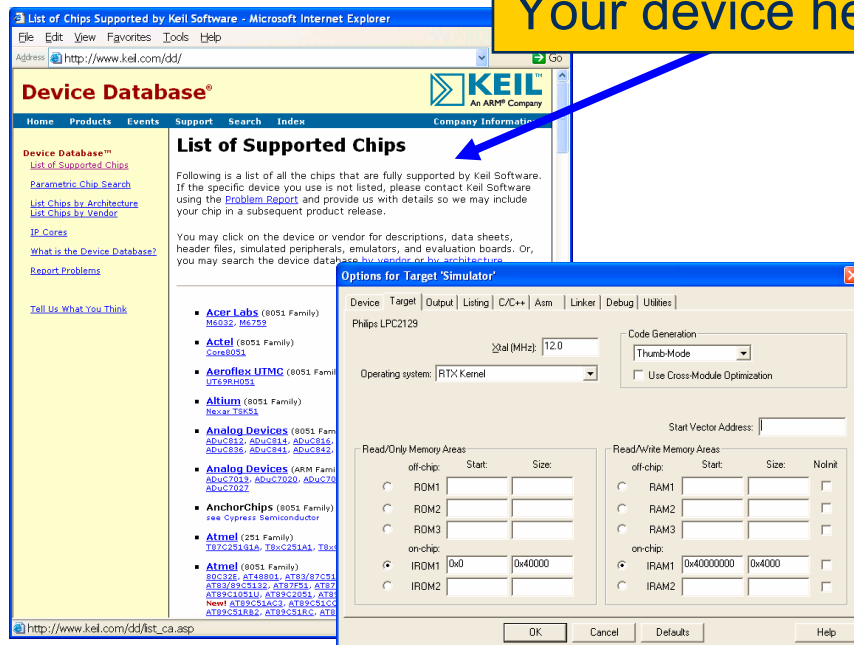
Step 2: Configure Device & Create Application Code



Device Database on Web



µVision includes Project Manager



Then Test Program

Step 3: Analyze Program with μ Vision Device Simulation

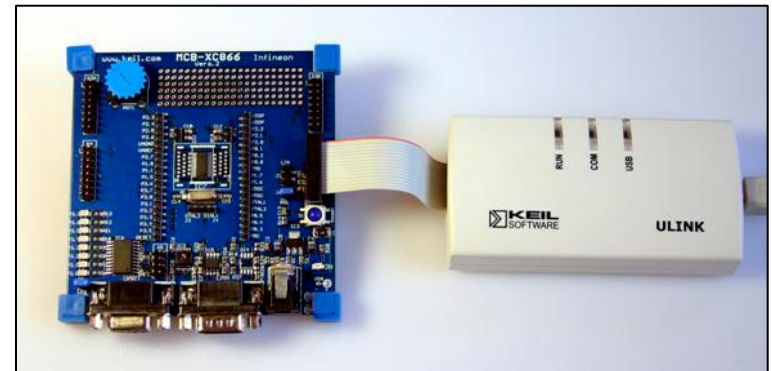
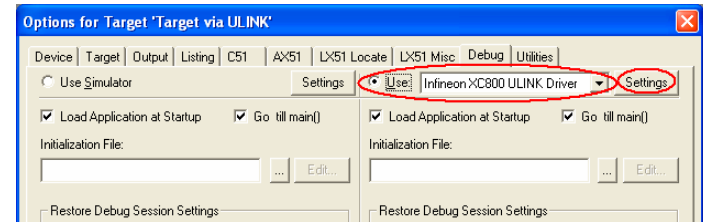
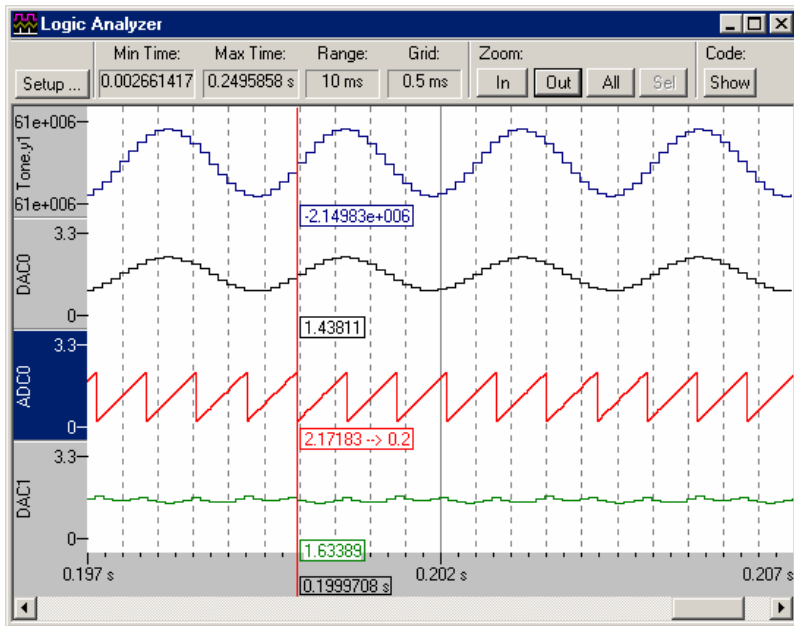
Step 4: Flash Download and Final Testing in Target Hardware



Debugging via μ Vision Debugger



Flash Download



Easy Mass Adoption

- One tool kit to support 150+ ARM processor-based devices
- Low touch technical support
- Global distribution network



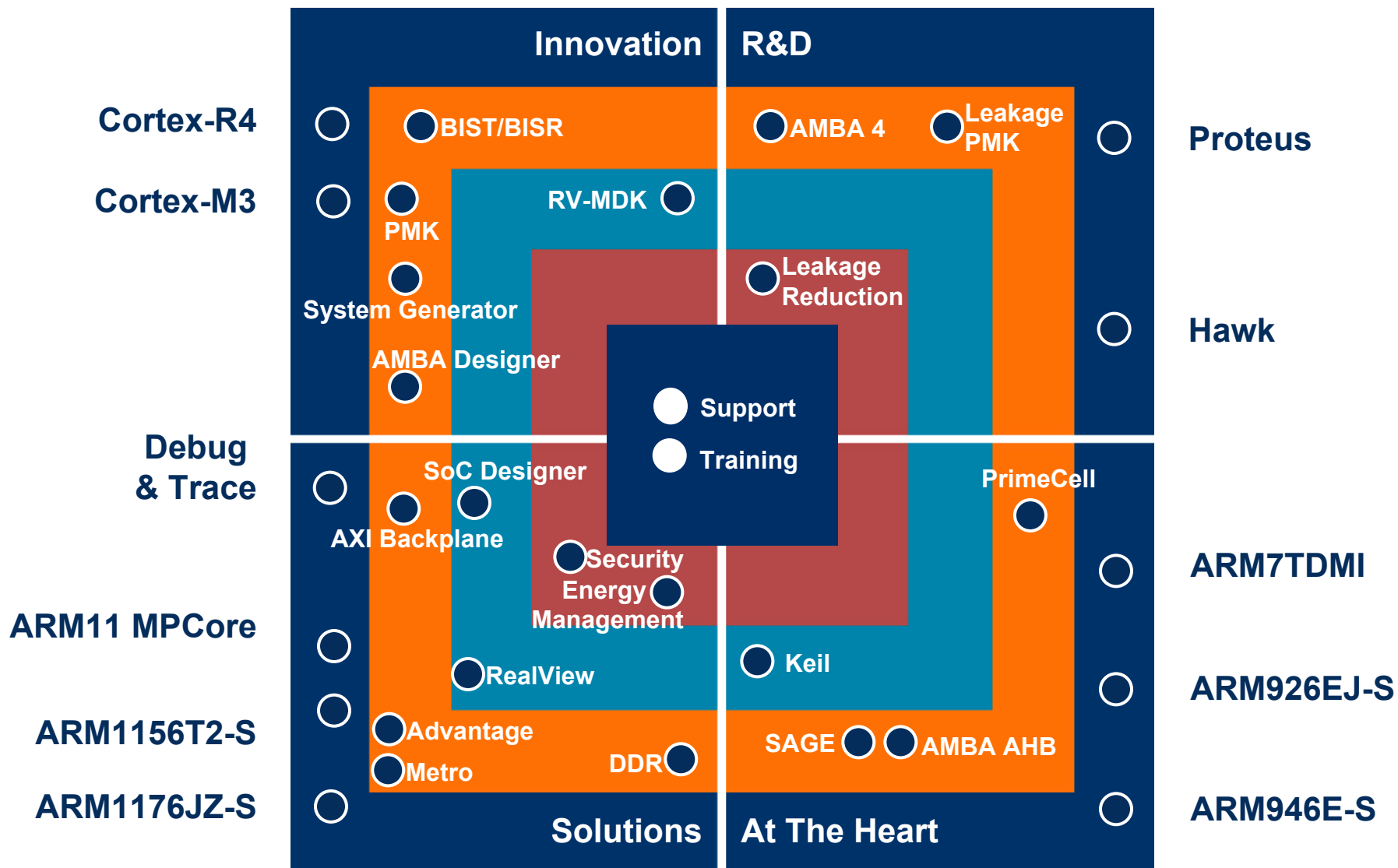
- Simple process for enabling your device
 - 2-12 weeks



MCU Starter Kit

- Benchmarking kit for evaluation
 - ISSM with cycle counting and RVDS 3.0 evaluation
 - CX model and SoC Designer evaluation
 - Download today from www.arm.com/support/downloads/index.html.
- Bundle for MCU development
 - Cortex-M3 license
 - RealView Development Suite
 - RealView ICE
 - FPGA board
 - Support added to RealView Microcontroller Development Kit

Embedded



Technical Seminars

- Introduction to Cortex-M3 processor
 - Tuesday, 2.45pm Fisher Auditorium
 - Wednesday, 4.00pm Castlereagh Room
- RealView Solutions for SoC Developers
 - Tuesday, 4.00pm Castlereagh Room
 - Thursday, 9.15am Castlereagh Room
- RealView Microcontroller Development Kit
 - Wednesday, 4.00pm Fisher Auditorium
 - Thursday, 10.30am Fisher Auditorium